

SocialCue: Exploring the Design Space of Social Wayfinding Assistants for Blind and Low Vision People

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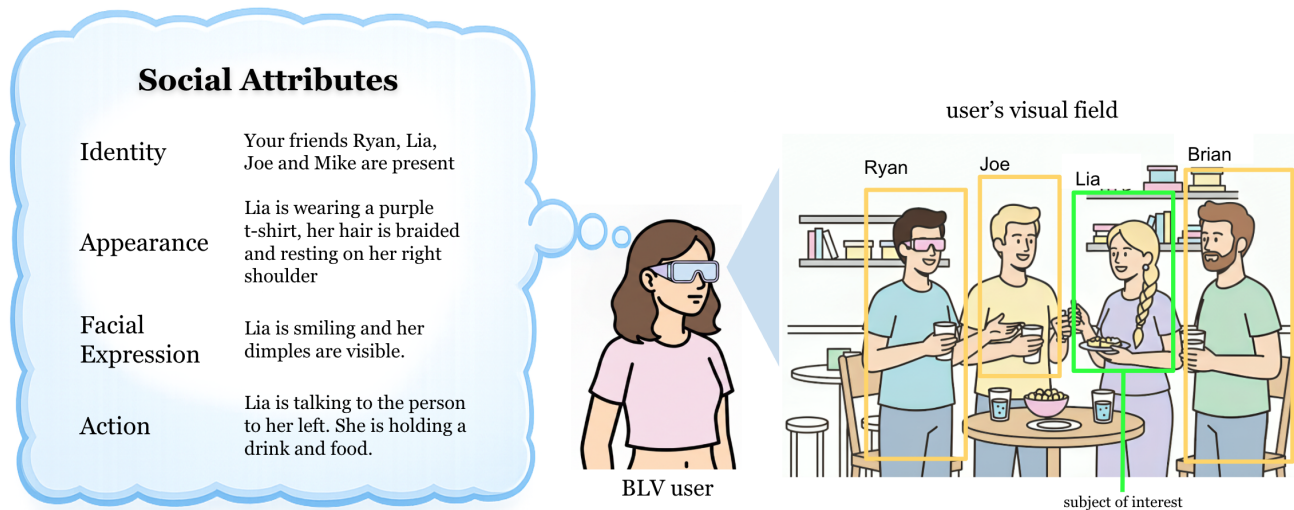


Figure 1: Overview of the SocialCue technology probe. A head-mounted system captures a blind or low vision (BLV) user's visual field in real time and uses computer vision to extract social cues from a scene, including other people's identity, facial expressions, actions, and physical details. SocialCue speaks these details via spatial audio to support social wayfinding.

Abstract

Social interactions often rely on the interpretation of visual cues. The inaccessibility of these nonverbal signals can relegate blind and low vision (BLV) people to the social periphery and restrict their independence in forming connections with others. Many assistive technologies for BLV people focus on spatial navigation or object recognition, but there is a gap in supporting "social wayfinding": the capacity to perceive and navigate human interaction dynamics. To address this, we explore the design space of a wearable system

that provides BLV users with real-time details about the social environment. The system, SocialCue, serves as a technology probe to explore BLV people's preferences for social navigation assistance. We conducted a two-phase formative study which identified four social attributes that the system should communicate: identity, social availability, facial expression, and physical descriptions including clothing and hairstyle. We describe the implementation of SocialCue and close by discussing our future evaluation plans.

CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools.**

Keywords

Social navigation, visual impairments, human detection, technology probe



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1 Introduction

Navigating a social environment is as complex as navigating physical space. For BLV individuals, the “cuelessness” caused by a lack of access to many nonverbal signals often leads to isolation and exclusion [12]. Sighted people rely on a constant stream of visual hints such as gaze and body language to regulate social interactions. Without these, BLV people must navigate social spaces with profound uncertainty, often having to manage potential embarrassment and social anxiety [5, 14, 16].

Consider Teng et al.’s example [14] of a blind person trying to orient themselves in a public space, using audio cues to find another person to talk to. He recounts the mortification of shouting, “Hey, where am I going?” only to realize he was trying to start a conversation with a garbage can. Although at the surface this can appear like just a navigational error, it reveals a deep social barrier. Such incidents can often force many BLV individuals into passive social roles, where they may stay silent to avoid the awkwardness of addressing the wrong person or missing a nonverbal turn-taking cue.

Current assistive technologies have made significant progress for supporting spatial independence and object identification. Systems like WorldScribe [3] provide live descriptions of static scenes, and tools like Microsoft’s PeopleLens [10] and Stearns and Thieme’s work [15] use audio to help users find specific peers in a room. From past research and our own interactions in mixed-ability teams, however, we have found that social “wayfinding” involves more than knowing where a person is or what objects are present [8, 14, 15]. It requires perceiving the dynamic signals that define human interaction: for example, is the person you are talking to smiling or looking away? Existing systems often fail to capture the nuances necessary for building rapport and not appearing like an “other.”

To bridge this gap, BLV people need explicit support for social navigation. Here, we explore the design space for this type of assistance by addressing the following research questions:

- RQ1.** What information about surrounding people do BLV people find most relevant for social wayfinding?
- RQ2.** How should assistive systems deliver social information to balance awareness, user agency, and cognitive load?

To answer RQ1, we conducted a two-phase formative study with BLV participants and found that beyond other people’s identity, users want access to nonverbal signals such as facial expressions and social availability. Additionally, participants expressed that physical descriptions, including hair color and clothing, were essential for re-identifying individuals and maintaining a consistent mental map of the social environment.

To answer RQ2, we developed SocialCue, a wearable technology probe that extracts and communicates these social attributes [7]. SocialCue delivers real-time details about the social environment

through spatialized audio and supports three interaction modalities. The first is “passive dwell,” where an automatic summary is triggered after the user holds their gaze on a person for three seconds. For deeper exploration, an “active query” allows users to pull specific details on demand. To prevent information overload, the system uses “progressive disclosure,” which provides increasingly detailed information if the user continues to focus on a person. SocialCue supports input through physical buttons, gestures and voice commands, providing a platform to study how different interaction methods impact a user’s sense of control and their ability to remain discreet in social settings. This work provides a foundation for establishing design guidelines for assistive technologies that support social inclusion. We close by discussing our future evaluation plans.

2 Related Work**2.1 Social challenges and interpersonal signals**

Research in interpersonal communication shows that navigating social interactions without sight requires managing challenges to one’s public image and avoiding potential stigma [12]. Thieme et al. describe how BLV individuals frequently negotiate their abilities to align with social norms and often have to balance the need for assistance with the desire for autonomy [15]. These social dynamics are complicated by the inaccessibility of nonverbal signals. For example, Qiu et al. note that the inability to perceive someone’s gaze direction and subtle gestures makes it difficult to regulate turn-taking and knowing who is being addressed. This lack of visual feedback often leads to feelings of social isolation and resorting to passive roles [11]. While prior work establishes the social consequences of inaccessible cues, a gap remains in translating these interpersonal needs into design requirements for real-time support. Our work aims to characterize BLV people’s needs and preferences in the context of assistive technology tailored to the problem of social “wayfinding”.

2.2 Wearable systems for social sensing

Prior research has tested various hardware to substitute for missing social signals, primarily through haptic feedback and head-mounted auditory displays. For example, Buimer et al. [2] used vibrotactile belts to convey six basic emotions detected from facial recognition software. By using haptic cues, this system is able to maintain the auditory channel open for conversation. While participants were able to detect the haptic signals, they identified wearability and robustness as important limitations of this approach. Similarly, Qiu et al. proposed wearable tactile bands that convey gaze direction to help users sense when they are being looked at and improve turn-taking dynamics [11].

Other approaches have used head-mounted displays to provide real-time audio feedback about social targets. One example is Microsoft’s PeopleLens [10], an AI system that uses spatial audio to read names and “orientation guides” to help users align their gaze with conversation partners. Similarly, Teng et al. [14] introduced the Person-Finder glasses, a HoloLens-based system that uses continuous audio cues to help BLV users locate and approach nearby strangers, aiming to reduce the social awkwardness of asking for help.

While these wearable systems support different aspects of social interactions, social wayfinding requires relaying both spatial information about surrounding people (where they are located with respect to the user) as well as social cues. We use SocialCue as a testbed for understanding how the two types of information should be designed and relayed. In our future evaluations with SocialCue, we will also explore design considerations for bystanders and for fostering effective interdependence [1] between BLV users, bystanders, and SocialCue itself. Lee et al. [9] found that bystanders are very accepting of wearable cameras when they know it is for accessibility purposes.

2.3 Managing information and context

Providing assistance with real-time descriptions of scenes requires balancing information richness with speed and cognitive load. WorldScribe [3] addresses this trade-off by using adaptive descriptions of the physical environment that change based on user behavior: it provides short labels during quick scans of a scene and includes more detail when the user is static.

Recent advancements in large language modeling (LLM), like ChatGPT's live video mode, demonstrate the potential for conversational interaction with surroundings. However, these chatbots often fail to describe dynamic events proactively, forcing users to ask repetitive questions [4]. Furthermore, using standard image captioning for social attributes raises ethical questions regarding accuracy and nuance. For example, Stangl et al. [13] found that preferences for this information are highly context-dependent; users may want detailed physical descriptions in dating scenarios but less in casual passing. With SocialCue, we follow these findings to build a configurable platform for studying when and how users prefer to access social attributes.

3 Formative Study

We conducted our formative research in two phases. In phase 1, we identified the social attributes that BLV people need about others in social situations, using static images of people in an elicitation study. In phase 2, we investigated additional design considerations the system must have to be useful in real-time interactions.

3.1 Phase 1: Identifying social attributes

We interviewed two BLV participants to understand which human-centric details they wanted in social images. Participants agreed that the identity of people around them was the most crucial factor for social awareness. Beyond identity, they wanted “visual confirmation” of nonverbal states. One participant noted they often feel “behind” in conversations because they cannot see facial expressions; knowing if someone is smiling or confused is essential for interpreting rapport. Finally, both participants wanted physical descriptions such as hair color and clothing as a functional tool to distinguish between people in a crowd. These findings led us to select four attributes: identity, social availability, facial expression and a physical description.

3.2 Phase 2: Additional design considerations for BLV users

We conducted co-design interviews with three BLV participants (P1–P3) to translate these attributes into a wearable system. We asked participants to describe social scenarios where they would find assistance to be helpful. We probed what their preferences would be for information delivery and modes of interaction with an ideal system. These sessions resulted in the following design requirements:

DR1: Re-identification through physical attributes: Participants noted that it is not enough to know that a person is nearby. P1 mentioned the value in “character-based” details “[...] *the kind of description that a book would give you if you're reading about a character, like hair type, color, if they're wearing glasses [...]* I think those basics should be the automatic description of the information that would allow you to leave the area and come back and reidentify the person regardless of where they are and what actions they're doing” (P1). Such physical attributes might enable users to track individuals as they move, ensuring a person remains a consistent “character” in their mental map even if their name is unknown.

DR2: Discretion and “hidden” interaction: P2 advocated for social inconspicuousness. He noted that hand gestures look “glitchy” to onlookers and requested physical buttons—on a device in a pocket or on the frame—to allow for “hidden” queries that do not signal the use of assistive technology. “*I would use it if it really didn't look like you had something on your head [...]* if it looked like a normal pair of sunglasses or a normal pair of glasses just with a little button on it” (P2).

DR3: Flexibility and noise control: P2 and P3 warned against a system that is too talkative. Referring to a constant data stream P3 expressed that “*if [the audio description] was on all the time I feel like that would be a little annoying*”. The system should allow the user to adjust information levels based on the environment or preference. “*It should be flexible [...] depending on the type of person and what their eye conditions are, they don't need certain things or they might need more things*” (P2).

3.3 Information Delivery Modes

Based on these phases, we identified three delivery modes for serving the different types of social information to users. In the spirit of research through design [6, 17], the system is configurable, allowing users to map each type of social information to one or more delivery modes so that they—and, in turn, our research team—can learn which options suit them best. The delivery modes can be toggled on or off in parallel.

- *Passive dwell:* The system provides an immediate spatial audio “ping” when a person is detected. If the user maintains their gaze on that person for 3 seconds, the system automatically announces a user-selected default attribute (typically identity or physical description), satisfying the need for effortless awareness.
- *Active query:* Users can “pull” specific details on demand. This allows for deep exploration without the system “pushing” unwanted data, addressing concerns about noise.
- *Progressive disclosure:* The system provides increasingly granular details—starting with a name, then an activity, then an

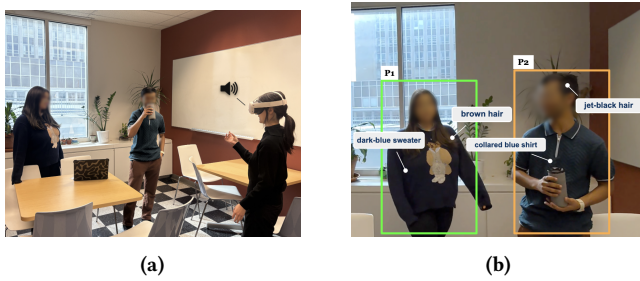


Figure 2: SocialCue system implementation and live feed. (a) A BLV user wearing a Meta Quest 3 headset to receive spatialized audio cues during a conversation. (b) A live feed from the system, demonstrating real-time person detection and the extraction of semantic attributes such as hair color and clothing, which SocialCue can then display to the user via one of its three delivery modes.

expression—if the user continues to point their head toward a person, mimicking natural gaze-based attention.

4 The SocialCue System

Our SocialCue technology probe is a distributed system with a Meta Quest 3 headset and a cloud-based computer vision backend for fast processing.

4.1 Camera Feed Processing Pipeline

The system streams visual frames from the headset to a cloud server, where four analysis modules operate in parallel.

Module 1: Identity detection. The first module identifies and tracks the physical location of people in the scene. It uses a YOLOv10 model that performs real-time human detection. The system then maps bounding box coordinates to the Quest’s 3D coordinate system to play spatialized audio using Head-Related Transfer Functions (HRTF).

Module 2: Physical description. In parallel, GPT-4o mini is prompted to extract a set of physical details that describe each person in the scene. This module generates a structured JSON profile per person that includes their clothing, whether they are wearing glasses, their hair color, hairstyle and eye color when visible. These physical attributes support re-identification over time.

Module 3: Action detection. To assess the action that each person in the scene is performing, the system uses a Vision Transformer (ViT) trained on human activity classification. This provides information such as “the person is looking at their phone,” which informs the BLV user about their social readiness.

Module 4: Facial expression. Finally, SocialCue includes a module for identifying the facial expression of each identified individual. This is achieved using DeepFace, a facial attribute analysis framework for Python.

4.2 User Interaction

The probe includes three input modalities. For discreet interaction, we use the Quest controller buttons as a proxy for physical buttons that could eventually be built into smart-glasses frames. This allows

users to trigger queries inconspicuously. For button-free use, we built a gesture-recognition pipeline using a finger-pinching system (e.g., thumb-to-middle finger for physical description, thumb-to-pinky for facial expression). Finally, we include the option for voice commands. Taken together, this approach allows us to systematically test for individual user preferences.

5 Conclusion and Future Directions

With SocialCue, our aim is to give BLV people explicit support for “social wayfinding” by surfacing nuanced social information of surrounding people and enabling BLV users to perceive this information in a flexible, user-driven way. By using an architecture with adjustable delivery “knobs,” the technology probe serves as a platform to further investigate how technology can support social agency for BLV people.

Our next step will be a set of field studies where BLV participants will use SocialCue in real time in diverse environments. Through these studies, we will undergo a research-through-design process [6, 17] (exploring different design configurations) to extract design principles for assistive technology that facilitates social wayfinding and contributes to inclusion. We will also explore design considerations for bystanders and for fostering effective interdependence [1] between all parties.

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